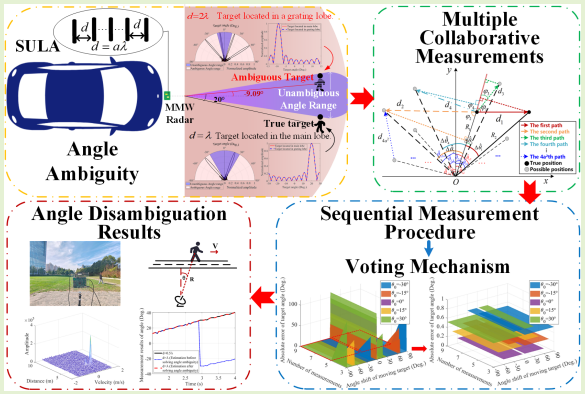
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Target Motion Enhanced Angle Estimation for Millimeter-Wave Radar with Sparse Uniform Linear Array

Jiangyou Zhu, Peng Lei, Jun Wang, Chi Zhang, Guidong He, and Mingming Jin

***Abstract*— A high angular resolution is crucial for refining the perception capabilities of millimeter-wave (MMW) radar for advanced driver assistant systems (ADASs). Sparse uniform linear arrays (SULAs) could serve as an effective and economical way for expanding array apertures to enhance the angular resolution. They provide analytical solutions while ensuring antenna continuity and mitigating the impact of mutual coupling. However, the presence of undesirable grating lobes in SULAs incurs angle ambiguity. Considering the movement of targets of interest in the sensed environment, this article proposes a novel angle-estimation method for MMW radar with SULA assisted by target motion information to address above issue. The grating-lobe characteristics and target angle ambiguity are theoretically analyzed to indicate their essential consistency. Based on this, multiple collaborative measurements are introduced to deduce all ambiguous angles. Additionally, their mutual-motion relationship imposes constraints on these ambiguous angles, promoting an initial angle estimation. A sequential measurement procedure and voting mechanism are applied to prevent nonunique solutions associated with target-angle shifts. These processes allow determining the final and unambiguous estimated angle, thereby enhancing the reliability and stability of the proposed method. Finally, its effectiveness is validated through numerical simulations and experiments.**



***Index Terms*— Angle estimation, sparse uniform linear array (SULA), millimeter-wave frequency modulated continuous wave (MMW FMCW) radar, moving target, ambiguous angle measurement.**

I. Introduction

ADVANCED driver assistant systems (ADASs) provide safe, comfortable, and efficient assistance functions in vehicles, attracting considerable attention [1]–[4]. Millimeter-wave (MMW) radars offer exceptionally precise object detection and localization capabilities, even in challenging environments, such as dense fog and dusty air, where the target is visually obscured [5]. Therefore, MMW radars are vital for environmental perception of ADASs [6], [7]. Most advanced MMW radar can simultaneously identify the distances, velocities, azimuths, and elevation angles of objects [8].

The angular resolution, which depends on the array aperture, is a major limitation to the MMW radar sensing [9]. Recently, taking advantage of multiple-input multiple-output (MIMO) techniques, the sparse linear arrays (SLAs) have gained popularity and attracted much attention in MMW radar systems. In contrast to traditional uniform linear arrays (ULAs), SLAs could produce a larger aperture to improve angular resolution with a given number of physical antenna elements [10]–[13]. Minimum redundant arrays (MRAs) [14], nested arrays (NAs) [15], and coprime arrays (CPAs) [16] are three basic structures of SLAs. They employ specific array constructions for increased angular resolution and additional degrees of freedom. To solve the array design problem, some optimization algorithms, such as particle swarm optimization [17]–[19], are explored to avoid high sidelobes and guarantee effective aperture with some stringent constraints and no analytical solutions [20]. Another type of SLAs is the sparse uniform linear arrays (SULAs), where physical antenna elements are uniformly located with the spacing much larger

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than a half-wavelength. Unlike intricate nonuniform arrays mentioned above, SULAs could provide analytical models for more concise array structure as well as more precise angle estimation, and also reduce system complexity and costs of MMW radars with high angular resolution [21].

However, SULAs suffer from undesirable grating lobes with the same amplitude as the main lobe. Besides energy dispersion and low gains, the angle ambiguity is a serious problem encountered in the application of MMW radars with SULAs. It has significant impact on the performance of target angle estimation [22]. Most existing studies to resolve the angle ambiguity in SLAs depend on some specific array geometries or the identification of ambiguity-free array structures [23]–[25], but is inapplicable to SULAs. In the case of high-precision ranging information and high signal-to-noise ratio (SNR) in MMW radars, the maximum likelihood and least squares methods are used to estimate target angle for SULAs in [26]. Based on the angular ambiguity function analysis, the alternating direction multiplier method in [27] allows the optimization of the sidelobe level for angle estimation in sparse array radar under some constraints of aperture size and minimum element spacing.

It should be noted that moving targets are of interest in MMW radar sensing, such as walking pedestrians in the automotive application. Therefore, different from those previous work, this article proposes a novel angle-estimation method for MMW radars with SULAs, which exploits the target motion information for addressing the angle ambiguity problem along with relaxing system and measurement constraints. First, the received signal model for SULA radars is established, and accordingly, a rigorous argument is presented to reveal the essential consistency between grating-lobe characteristics and target angle ambiguity in SULAs. On the basis, a multiple-collaborative measurement scheme is introduced to identify the potential angle of radar target from ambiguous estimates. In addition, we find and explicitly prove that there exist nonunique solutions in the scheme arising from target angle shifts at some specific angles. Thus, a sequential measurement procedure and voting mechanism is further developed to overcome such problem. Finally, numerous simulations and experiments are carried out to illustrate the effectiveness of the proposed method for target angle estimation in the MMW radar with SULA. The main contributions of this work could be summarized as follows:

(1) Target motion information and multiple collaborative measurements are introduced to address the angle-ambiguity issue based on mutual coupling constraints for MMW radar with SULA.
(2) A sequential measurement procedure and voting mechanism are organically integrated to overcome the nonunique solution problem of target-angle shifts in the angle-estimation method.
(3) The proposed method is compatible with SULA, regardless of antenna spacing, thereby simplifying signal processing and enhancing its applicability.

The rest of this article is organized as follows. Section II investigates the essential characteristics of target angle ambiguity for the MMW radar with SULA. The details of proposed method is presented in Section III. Simulations and experiments are performed in Section IV to validate the proposed method. Section V gives the conclusion.

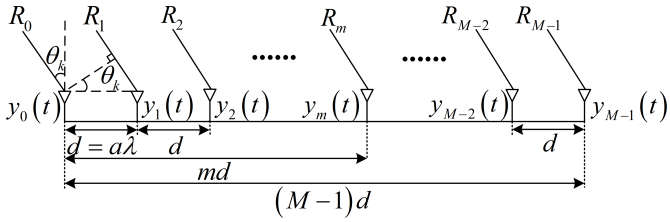


Fig. 1. Illustration of SULA.

II. Angle Ambiguity Analysis in SULA Radar

A. Signal Model

Consider a MIMO radar system with transmitting antennas (M_t) and receiver antennas (M_r) , where the total number of receiver antennas is M, which is the product of M_t and M_r. Additionally, the number of targets is K, and the MMW radar employs the FMCW modulation scheme. An illustration of SULA is shown in Fig. 1, where d denotes the distance between two adjacent antennas ($d = a\lambda, a > 0.5$), θ_k is the incidence angle of the k-th target ($k \in [1, K]$, $\theta_k \in [-0.5\pi, 0.5\pi]$), λ is the signal wavelength, R_m is the distance between the far-field signal wavefront and the m-th antenna $(m \in [0, M-1])$, $y_m(t)$ is the signal received by the m-th antenna, and the first antenna is the reference antenna $(m = 0)$. The differences in the signals received by each antenna are exclusively caused by phase differences. Therefore, the signal received by the m-th antenna from the k-th target is expressed as

$$\begin{aligned} s_k(t-\tau_{mk}) &= s_k(t) \cdot e^{-j2\pi f_0 \tau_{mk}} \\ &= s_k(t) \cdot e^{-j\frac{2\pi}{\lambda} md \sin\theta_k} \end{aligned} \tag{1}$$

where $s_k(t)$ is the signal received by reference antenna from the k-th target, f_0 is the carrier frequency, $\tau_{mk} = md\sin\theta_k/c$ is the delay of the k-th target signal arriving at the m-th antenna, and c is the speed of light. Therefore, the signal received by the m-th antenna from all K targets is

$$y_m(t) = \sum_{k=1}^{K} \left[b_{mk} \cdot s_k(t-\tau_{mk}) + n_{mk}(t) \right] \tag{2}$$

where b_{mk} and $n_{mk}(t)$ denote the gain and noise of the m-th antenna for the k-th target, respectively. The signal received by the m-th antenna at the n-th snap is

$$y_{mn} = \sum_{k=1}^{K} \left[b_{mk} \cdot s_k(t_n) \cdot e^{-j\frac{2\pi}{\lambda} md \sin\theta_k} + n_{mk}(t_n) \right] \tag{3}$$

Assuming that the number of snaps is N, we obtain

$$\mathbf{y_m} = \left[y_{m1}, \cdots, y_{mn}, \cdots, y_{mN} \right]^T \tag{4}$$

$$\mathbf{s_k} = \left[s_{k1}, \cdots, s_{kn}, \cdots, s_{kN} \right]^T \tag{5}$$

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The signals received by the M antennas at all N snaps are arranged in a matrix, yielding the following signal model:

$$\mathbf{Y} = \mathbf{AS} + \mathbf{N} \tag{6}$$

where $\mathbf{Y} = [\mathbf{y_0}, \cdots, \mathbf{y_m}, \cdots, \mathbf{y_{M-1}}]^T \in \mathbb{C}^{M\times N}$ denotes the received-signal matrix, $\mathbf{A} = [\mathbf{a_1}, \cdots, \mathbf{a_k}, \cdots, \mathbf{a_K}] \in \mathbb{C}^{M\times K}$ is the array-manifold matrix, $\mathbf{S} = [\mathbf{s}_1, \cdots, \mathbf{s_k}, \cdots, \mathbf{s_K}]^T \in \mathbb{C}^{K\times N}$ is the source-signal matrix, $\mathbf{N} \in \mathbb{C}^{M\times N}$ is the noise matrix, and $\mathbf{a_k}$ is the steering vector. Under ideal conditions, each antenna is isotropic; thus, b_{mk} can be normalized to one. Therefore, $\mathbf{a_k}$ can be expressed as

$$\mathbf{a_k} = \left[e^{-j2\pi f_0 \tau_{0k}}, \cdots, e^{-j2\pi f_0 \tau_{mk}}, \cdots, e^{-j2\pi f_0 \tau_{(M-1)k}}\right]^T \tag{7}$$

B. Analysis of Angle Ambiguity in SULA Based on Grating-Lobe Characteristics

1) Analysis of Grating-lobe Characteristics: The phase of a SULA changes uniformly, which is defined as α. The array factor [22] is expressed as

$$S(u) = \sin\left(\frac{Mu}{2}\right), u = \frac{2\pi}{\lambda} d \sin\theta_k + \alpha \tag{8}$$

Maximum values occur periodically when $u = 2\pi i, i = 0, \pm1, \pm2, \ldots$. When $i = 0$, the angle corresponding to the maximum value is θ_k, i.e., the target angle is located in the main lobe. When $i \neq 0$, the angle corresponding to the maximum value is θ_s, i.e., the target angle is located in the grating lobe, which can be expressed as

$$\begin{aligned} \theta_s &= \sin^{-1}\left(\frac{(2\pi i - \alpha)\,\lambda}{2\pi d}\right) \\ &= \sin^{-1}\left(\frac{i}{a} + \sin\theta_k\right), i = \pm1, \pm2, \ldots \end{aligned} \tag{9}$$

According to the properties of the inverse sine function, the condition satisfied by i is expressed as

$$a\left(-1 - \sin\theta_k\right) \leq i \leq a\left(1 - \sin\theta_k\right) \tag{10}$$

In particular, for a side-scan array, i.e., $\theta_k = 0°, \alpha = 0$, the condition satisfied by i is

$$-a \leq i \leq a, i = \pm1, \pm2, \ldots \tag{11}$$

Equivalence on both sides of the inequality cannot be obtained simultaneously. Therefore, the number of possible values for i is $2a$, i.e., the total number of main and grating lobes can be expressed as

$$p = 2a \tag{12}$$

2) Angle-ambiguity Analysis: The signal received from the k-th target can be expressed as

$$y[m] = s_k(t) \cdot e^{-j\frac{2\pi}{\lambda} m d \sin\theta_k} \tag{13}$$

Assuming that the spectrum of $s_k(t)$ is $X(e^{j\omega})$, the spectrum of $y[m]$ can be expressed as

$$Y\left(e^{j\omega}\right) = X\left(e^{j(\omega-\omega_0)}\right), \omega_0 = -\frac{2\pi d \sin\theta_k}{\lambda} \tag{14}$$

As $\theta_k \in \left[-0.5\pi, 0.5\pi\right]$ and $\omega_0 \in 2a[-\pi, \pi]$, the number of discrete-time Fourier transform (DTFT) spectrum aliasing occurrences is

$$q = 2a \tag{15}$$

SULA does not satisfy the Nyquist sampling theorem [28] in the spatial domain because its antenna spacing is larger than half-wavelength, which leads to spectrum aliasing. Additionally, spatial undersampling appears as grating lobes in the SULA beam pattern and as new frequency components in the SULA spectrum. By analyzing these two theoretical perspectives, we found that $p = q$. Thus, the intrinsic consistency between grating-lobe characteristics and target angle ambiguity effect was demonstrated.

Assuming that the number of discrete Fourier transform (DFT) is L, the signal spectrum is

$$Y[l] = Y\left(e^{j\omega}\right)\Big|_{\omega=\frac{2\pi}{L}l} = X\left(e^{j\left(\frac{2\pi}{L}l - \omega_0\right)}\right), l \in [0, L-1] \tag{16}$$

In particular, when $\omega - \omega_0 = 0$, we obtain

$$\omega = \omega_0 = -2\pi a \sin\theta_k = \frac{2\pi l}{L} \tag{17}$$

Therefore, the unambiguous angle range, defined as Ω, is expressed as

$$\theta_k \in \left(\sin^{-1}\left(-\frac{1}{2a}\right), \sin^{-1}\left(\frac{1}{2a}\right)\right] \tag{18}$$

When $\omega - \omega_0 = 2\pi i, i = 0, \pm1, \pm2, \ldots$, the maximum values of this spectrum are obtained. The DFT number corresponding to the maximum value can be expressed as

$$l = \frac{(\omega_0 + 2\pi i)\,L}{2\pi} = \left(-\frac{d\sin\theta_k}{\lambda} + i\right) L \tag{19}$$

Therefore, the angle estimate of SULA is

$$\hat{\theta}_k = \sin^{-1}\left(-\frac{\lambda}{d}\left(\frac{l}{L} - i\right)\right) = \sin^{-1}\left(\frac{i}{a} + \sin\theta_k\right) \tag{20}$$

SULAs have only one main or grating lobe within their unambiguous angle range. Therefore, we obtain

$$\hat{\theta}_k = \begin{cases} \theta_k, & i = 0 \\ \sin^{-1}\left(\frac{i}{a} + \sin\theta_k\right) = \theta_s, & i = \pm1, \pm2, \ldots \end{cases} \tag{21}$$

The unambiguous angle range is $(-30°, 30°]$ when $d = \lambda$. The estimated angles between $-60°$ and $60°$ compared to the full ULA, i.e., $d = 0.5\lambda$, are shown in Fig. 2.

Theorem 1 (Angle uniqueness)**.** *SULA has only one main or grating lobe in the unambiguous angle range.*

Proof: See Appendix I.

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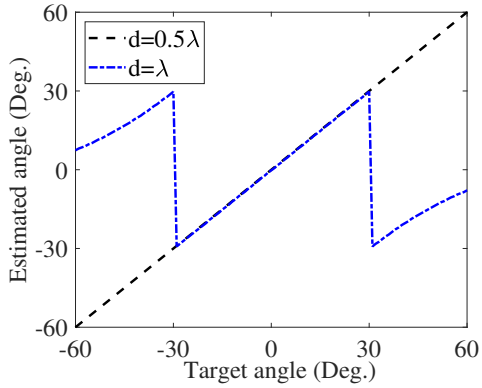


Fig. 2. Estimated angle versus target angle.

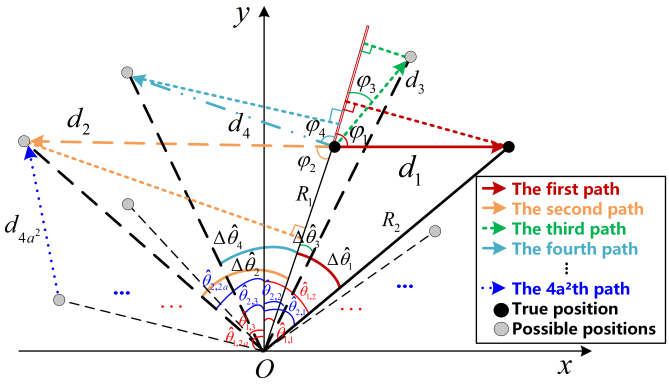


Fig. 3. All possible paths of the moving target.

III. Proposed Method

The theoretical analyses presented in Subsection II-B indicate that expanding aperture using SULA introduces angle ambiguity. To overcome this problem, we propose an angle-estimation method for an MMW radar with SULA assisted by the motion information of target.

First, the target distance, radial velocity, and angle are estimated through three-dimensional Fast Fourier Transform (3D FFT) [29]. SULA only introduces angle ambiguity without affecting the estimated distance or radial velocity [30]. Second, all ambiguous angles were derived according to (21). Third, based on the estimated distance and all possible angles, all potential locations where the target may appear were determined. Fourth, all possible motion paths of the target were derived, followed by backtracking to deduce the radial velocities that correspond to each possible path. Matching these with the measured radial velocity allowed for the identification of the correct motion path and target angle. Fifth, a sequential measurement procedure and voting mechanism are proposed to overcome nonuniqueness in target-angle shifts at specific angles. The final target angle was determined by selecting the angle that appeared most frequently in a series of sequential measurements. These enhancements improved the reliability and stability of our method.

A. Geometry of Moving Target using MMW Radar with SULA

The ambiguity of angle results in multiple possible motion paths. All possible paths of the target in the Cartesian coordinate system are shown in Fig. 3, where the red solid line represents the true path of the target and the dotted lines represent all possible paths. We assumed that solid red line represents the positive direction. R_1 and R_2 denote the distances between radar and target for the first and second measurements, respectively. Similarly, $\hat{\theta}_{1,x}, x \in [1, 2a]$ and $\hat{\theta}_{2,x}$ indicate the possible angles for the first and second measurements, respectively. $\Delta\hat{\theta}_y, y \in [1, 4a^2]$ is the angle shift corresponding to the y-th path, d_y is a possible path, and the angle of the radial velocity projection is expressed as

$$\varphi_y = \sin^{-1}\left(\frac{R_2 \cdot \sin\Delta\hat{\theta}_y}{d_y}\right) \tag{22}$$

B. Doppler Velocity Estimation via Range and Angle Extraction

The velocity corresponding to d_y is calculated as

$$\nu_y = \frac{d_y}{t} = \frac{\sqrt{R_1^2 + R_2^2 - 2R_1 \cdot R_2 \cdot \cos\left(\Delta\hat{\theta}_y\right)}}{t} \tag{23}$$

where t denotes the motion time and can be expressed as

$$t = \mathrm{PRT} \cdot \mathrm{numChirps} \tag{24}$$

where PRT is the pulse repetition time and numChirps is the number of linear frequency modulation pulses.

Accordingly, the radial velocity corresponding to d_y is

$$\nu_{dy} = \nu_y \cdot \cos\varphi_y = \frac{\sqrt{d_y^2 - \left(R_2 \cdot \sin\Delta\hat{\theta}_y\right)^2}}{t} \tag{25}$$

Finally, we reverse-deduced the velocities and radial velocities corresponding to all possible paths.

C. Angle Disambiguation

Assume that the initial angle of moving target is θ_0, and its radial velocities for the first and second measurements are V_1 and V_2, respectively. Thus, the angle ambiguity can be addressed by solving the following optimization problem:

$$\mathrm{P}_y = \mathrm{argmin}\left\{\left[\left|V_1 - \nu_{dy}^1\right| + \left|V_2 - \nu_{dy}^2\right|\right]\right\} \tag{26}$$

where ν_{dy}^1 and ν_{dy}^2 denote the possible radial velocities of the target for the first and second measurements, respectively. Ultimately, we determined the angle corresponding to P_y with the minimal absolute error as the resolved angle ambiguity.

The angle disambiguation process described above is called multiple collaborative measurements, which use mutual constraints among distances, velocities, and all possible angles before and after the movement of the target. It enables accurate angle estimations through a decision criterion of minimizing the sum of the absolute values of radial velocity errors. The interrelationships between all parameters are shown in Fig. 4. Firstly, we measured the distance, radial velocity, and angle of the target twice before and after its motion. Secondly, the distance and angle were used to calculate the possible positions of the moving target. Thirdly, all possible motion

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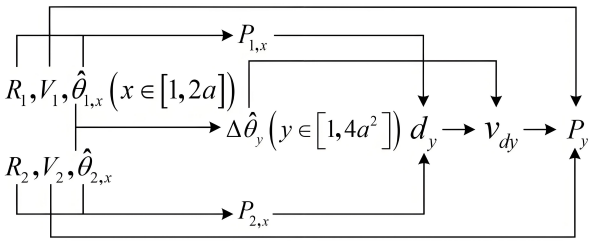
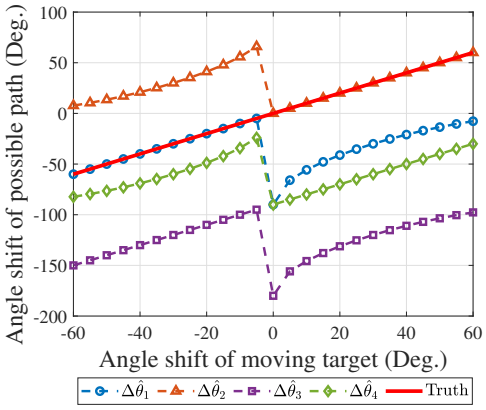
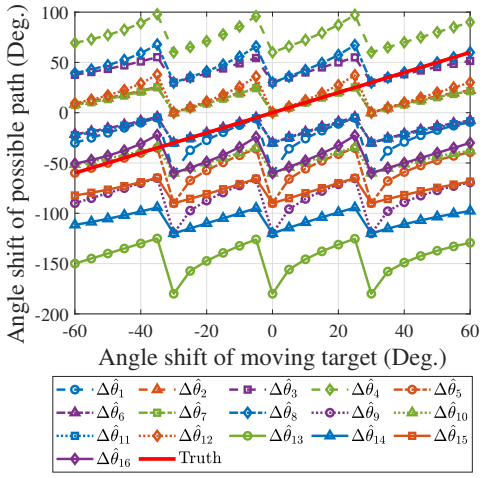


Fig. 4. Constraint relationships among the multipoint collaborative measurement parameters.



(a)



(b)

Fig. 5. Angle shift of possible path versus angle shift of target. (a) $d = \lambda$. (b) $d = 2\lambda$.

paths of the target were obtained. Fourthly, these motion paths were combined with their corresponding angle shifts to derive the corresponding radial velocities. Finally, based on all previously calculated radial velocities and the true measured radial velocity, a decision criterion that minimizes the absolute error was applied to confirm the true motion path of the target, thus determining its true angle.

In summary, although we addressed the angle-ambiguity problem, nonunique angle shifts at special angles arises in our method. They range between $[-60°, 60°]$ when $\theta_0 = 0°$, and the angle shifts corresponding to all possible paths are shown in Fig. 5. When the target moves to a certain angle shift,

Algorithm 1 Target Motion Enhanced Angle Estimation for Millimeter-Wave Radar with Sparse Uniform Linear Array

Input: Multiple frames of radar echo data
Output: Angle estimates $\hat{\theta}_k$
Initialization: $d = a\lambda, a > 0.5$, optimal number of observations Z and fixed time ΔT;
for $z \leftarrow 1$ **to** Z **do**
  Determine the value of z;
  if $z < Z$ **then**
    Extract single frame according to ΔT;
    Calculate R_z and V_z;
    Calculate $\hat{\theta}_{z,x}$ based on (21);
    Calculate all possible positions of the target $\left(R_z \sin\left(\hat{\boldsymbol{\theta}}_{z,x}\right), R_z \cos\left(\hat{\boldsymbol{\theta}}_{z,x}\right)\right)$;
    Calculate all possible velocities based on (23);
    Calculate ν_{dy}^z corresponding to ν_y^z based on (25);
    Calculate $\mathbf{P}_y^z$ using based on (26);
    Calculate the target angle $\hat{\theta}_k^z$ corresponding to $\mathbf{P}_y^z$;
    $z \leftarrow z + 1$;
  else
    Use the voting mechanism to determine ultimate target angle $\hat{\theta}_k$ from all sequential measurement results $\hat{\theta}_k^z$;
  end
end

there are multiple intersection points between the true and all possible values of $\Delta\hat{\theta}_y$, indicating that the solution of the proposed method is not unique as antenna spacing increases.

Theorem 2 (Nonunique angle shifts)**.** *At some specific angles, the angle shifts are not unique.*

Proof: See Appendix II.

To address the nonuniqueness issue, we further implemented a sequential measurement procedure inspired by [31]. The core concept of sequential measurements is to perform multiple collaborative measurements sequentially to obtain the target signals and use them to derive the target angle. Compared to traditional single measurements, sequential measurements can provide richer and more accurate information, thereby facilitating accurate target angle estimations. Subsequently, (26) was employed to determine the optimal path for each measurement and obtain the corresponding angles.

Suppose that the number of optimal measurements is Z and the angle estimated from the z-th measurement is $\hat{\theta}_y^z$, which corresponds to the minimum absolute error. The core idea of the voting mechanism is to vote on a set of candidates and select that with the highest number of votes. This essentially represents a decision-making method based on statistical principles. Therefore, we utilized a voting mechanism that relies on the majority to determine the final angle from Z angles.

In conclusion, we employed the pseudocode presented in Algorithm 1 to articulate the process of the proposed method. Additionally, we evaluated the computational complexity of this algorithm, which primarily comprises two parts: 3D FFT and angle unambiguity. It can be approximately expressed as

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TABLE I

MMW RADAR PARAMETER SETTINGS IN SIMULATIONS AND EXPERIMENTS

Parameters	Values	Parameters	Values
Carrier frequency	77 GHz	Modulation slope	58 MHz/μs
Bandwidth	2.97 GHz	Sampling rate	20 MHz
PRT	$7.81 \times 10^{-4}\,s$	Frame duration	$0.05\,s$
Range FFT points	4096	Doppler FFT points	256
Number of sampled points per chip	512	Number of chirps per frame	64

$$\mathcal{O}\left[Z\left(\underbrace{N_R N_V \log(N_R N_V) + L \log L}_{\text{3D FFT}} + \underbrace{4a^2 N_M}_{\text{angle unambiguity}}\right)\right] \tag{27}$$

where N_R, N_V, and L indicate the numbers of FFT for the distance, radial velocity, and angle, respectively. N_M is the computational complexity of single measurement, which is considerably lower than those of N_R, N_V, and L.

For a small a, the computational complexity is primarily determined via 3D FFT. As a increases, the angle unambiguity becomes dominant. Using Algorithm 1, we addressed the problem of angle ambiguity and enhanced the robustness and effectiveness of MMW radar with SULA for target angle estimation. These advancements significantly reduce costs and enhance the performance of autonomous vehicles.

IV. Results and Discussion

The parameters of MMW radar are set to be the same in both simulations and experiments of this section for the consistency of performance evaluation, as listed in Table I. To ensure that the number of antennas is an integer for $d = \lambda$ and $d = 2\lambda$, we assume that array aperture is 40λ. The time of one frame is $0.05\,s$, the velocity is $1\,m/s$, Z is the most optimal number of measurements for sequential measurements, and $\Delta\theta$ is the most optimal angle shift. Additionally, the distance and velocity resolutions are $0.05\,m$ and $0.04\,m/s$, respectively.

A. Simulation Results

1) Influence of Grating Lobes in SULA: The number of antennas for $d = \lambda$ and $d = 2\lambda$, is 40 and 20, respectively. The beam patterns are shown in Fig. 6, wherein one grating lobe appears in the beam pattern for $d = \lambda$ and the estimated angle exhibits two ambiguous values. The main and grating-lobe angles for $\theta_0 = 0°$, are $0°$ and $90°$, respectively, whereas they are $41.15°$ and $20°$ for $\theta_0 = 20°$, respectively. Similarly, when $d = 2\lambda$, three grating lobes appear, resulting in four ambiguous values. Additionally, for $\theta_0 = 0°$, the ambiguous values are $-30°$, $0°$, $30°$, and $90°$, where $0°$ represents the main lobe, whereas the others represent the grating lobes. When $\theta_0 = 20°$, the four ambiguous values are $-41.15°$, $-9.09°$, $20°$, $57.35°$, where $20°$ represents the main lobe and the others represent grating lobes. In short, as d increases, the number of grating lobes gradually increases compared to the original array. The above analyses precisely correspond to the grating-lobe characteristics and angle-ambiguity effect discussed in Subsection II-B.

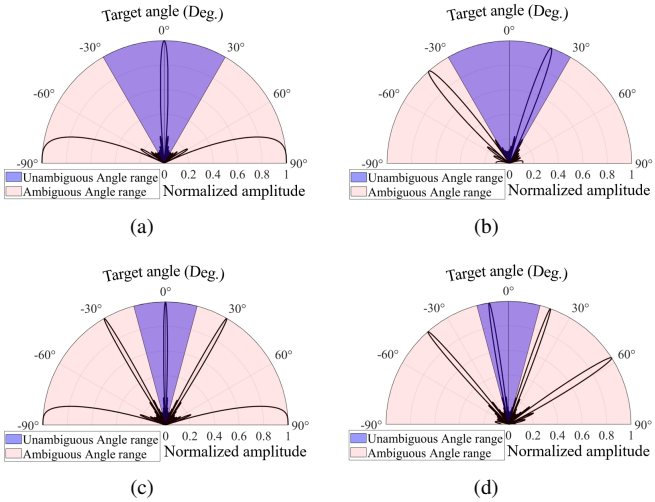


Fig. 6. Beam patterns of SULA. (a) $d = \lambda, \theta_0 = 0°$. (b) $d = \lambda, \theta_0 = 20°$. (c) $d = 2\lambda, \theta_0 = 0°$. (d) $d = 2\lambda, \theta_0 = 20°$.

The angular spectra of targets located in the main and grating lobes for $d = \lambda$ and $d = 2\lambda$ are shown in Fig. 7(a) and Fig. 7(b), respectively. When $d = \lambda$, the unambiguous angle ranges between $(-30°, 30°]$, and the estimated angle of target located in the main or grating lobes is $20°$. However, when $d = 2\lambda$, the unambiguous angle ranges between $(-14.48°, 14.48°]$; in all cases, the estimated angle is $-9.09°$. Therefore, if the target is in the main or grating lobe, the estimated angle is within unambiguous range, corresponding to the results shown in Fig. 6(b), Fig. 6(d) and the analyses in Subsection II-B.

2) Target Angle Estimation with Respect to Angle Shifts and Initial Angle: The relationships among Z, $\Delta\theta$, and the absolute error of the target angle for $d = \lambda$ and $d = 2\lambda$ are shown in Fig. 8. The left-hand side panels illustrate the relationships among $Z \in [3, 9], \Delta\theta \in [-60°, 60°]$ and absolute error, whereas the right-hand side panels are enlarged versions of the left, indicating that the estimated angles remain unaffected by special angles, thereby ensuring their uniqueness.

Given that the angle shifts are nonunique at special angles, each individual $\Delta\theta$ leads to an error, resulting in dramatic peaks. Therefore, Z should be increased and set to an odd value as voting mechanism relies on majority.

The absolute errors were less than $1°$ for $d = \lambda$, $Z = 3$, $\Delta\theta = 3°$ and $d = 2\lambda$, $Z = 5$, $\Delta\theta = 5°$. Therefore, the optimal Z and $\Delta\theta$ could be determined using the sequential measurement procedure and voting mechanism. Note that Z should be an odd number, as it ensures that there always exists a definitive majority in the voting mechanism. The relationships among Z, $\Delta\theta$, and the angle absolute error of the target angle were analyzed as follows:

(1) As the angle increased, the absolute error also increased, which is consistent with the radar-angle accuracy trend.
(2) For $\theta_0 = 0°$, regardless of $\Delta\theta$, the absolute angular errors remained the same owing to the symmetry.
(3) For $\theta_0 \neq 0°$, the absolute angular errors exhibit symmetry

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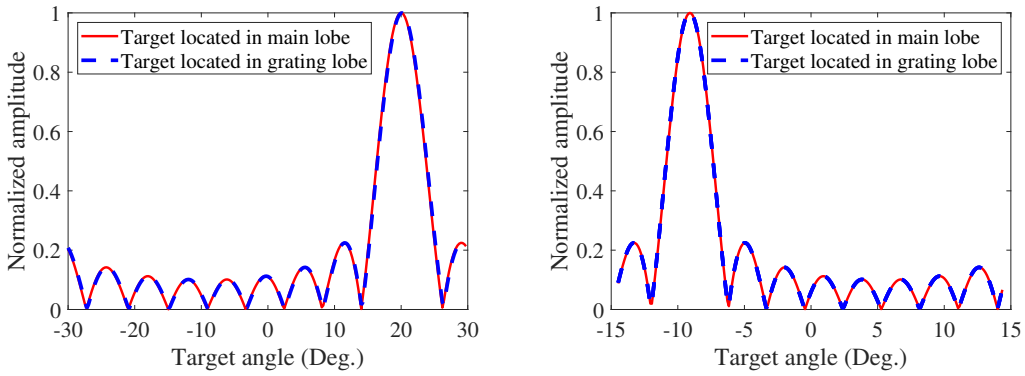


Fig. 7. Angular spectra for target located in main and grating lobes. (a) $d = \lambda$. (b) $d = 2\lambda$.

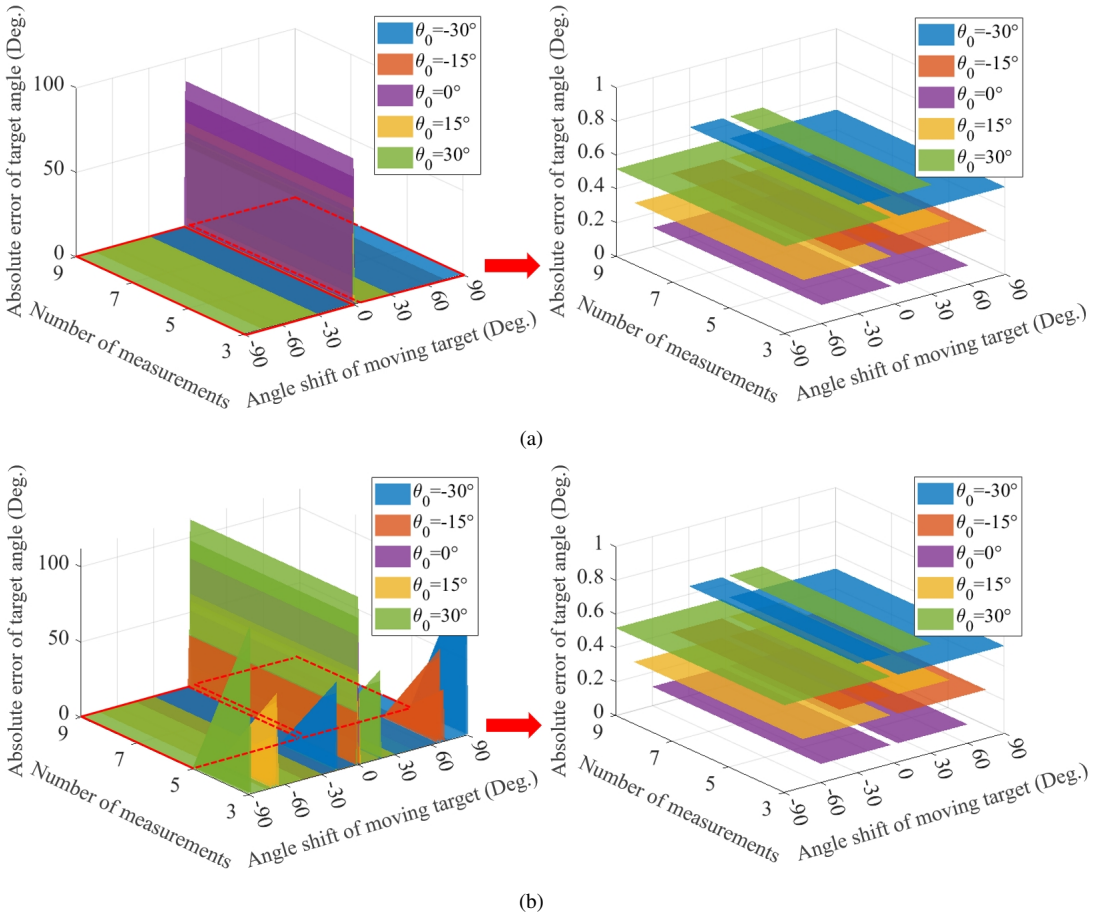


Fig. 8. Relationships among Z, $\Delta\theta$, and absolute angular error. (a) $d = \lambda$. (b) $d = 2\lambda$.

when targets initially positioned at symmetric angles (θ_0 and $-\theta_0$) move in opposite directions. Given that the center of SULA is at its origin, a conjugation relationship exists between the two signals, resulting in opposite estimates. Consequently, the absolute error are equal for two targets with symmetric initial angles and moving in opposite directions.

(4) For $\theta_0 \neq 0°$, when sharing the same initial angle but moving in opposite directions, the absolute angular errors become asymmetric. This is because there is no signal conjugation between them, and then affect the final angle estimates. Consequently, the absolute angular errors are not equal for two objects with identical initial angles that move in opposite directions.

The absolute angular errors versus the target angles for $d = \lambda$, $d = 2\lambda$, and $d = 4\lambda$ are shown in Fig. 9. Based on optimal Z and $\Delta\theta$, the sequential measurement procedure and voting mechanism resulted in symmetrical absolute errors. Additionally, as the target moved in the positive direction, the absolute errors lacked a strict symmetry. Nevertheless, the

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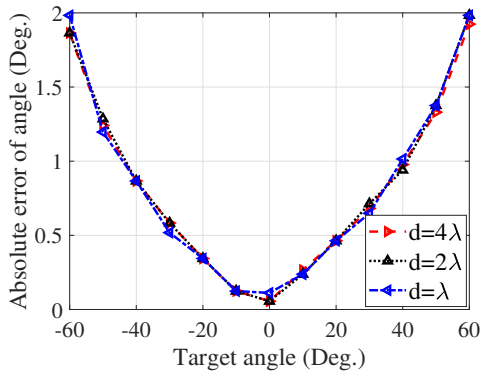


Fig. 9. Absolute target angular errors versus target angles.

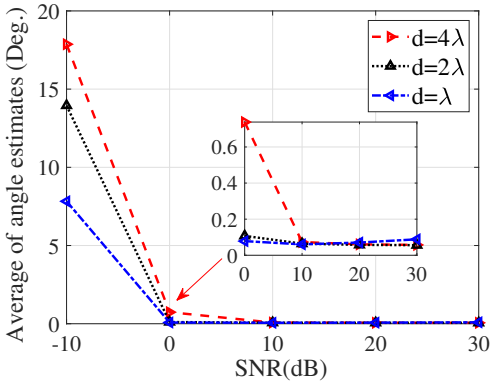


Fig. 11. Mean absolute target angular errors versus SNR ($\theta_0 = 0°$).

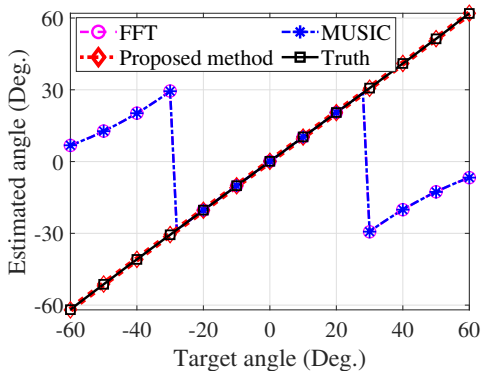


Fig. 10. Comparison of different angle estimation methods for SULA radar ($d = \lambda$).

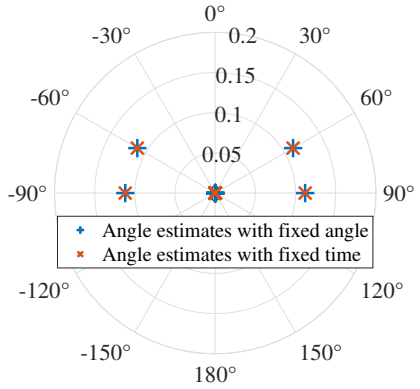


Fig. 12. Absolute angular errors divided by fixed time and fixed angle ($\theta_0 = 0°$).

deviation was small and its impact was subtle. It should also be noted that different target angles off the SULA normal direction also influence unambiguous angle estimates for the moving target owing to the change of beam width.

Furthermore, Fig. 10 compares angle estimates for the given SULA radar with $d = \lambda$ using FFT, MUltiple SIgnal Classification (MUSIC) [32] and the proposed method, respectively. It can be seen that all three methods achieve accurate estimation in close agreement with corresponding true values in the unambiguous angle range $(-30°, 30°]$. However, in the ambiguous angle ranges $[-60°, -30°]$ and $(30°, 60°]$ of the SULA, the proposed method could enable the unambiguous angle estimation, while FFT and MUSIC based counterparts suffer from estimation errors caused by grating lobes.

3) Target Angle Estimation with Respect to SNR: The absolute angular errors versus SNR (after pulse compression) for $\theta_0 = 0°$ are shown in Fig. 11. We conducted 100 Monte Carlo simulation for $d = \lambda$, $d = 2\lambda$, and $d = 4\lambda$. As the SNR increased from -10 to 30 dB, the absolute angular error decreased. Here, we can also observe that the antenna spacing exerts an impact on angle estimates of the moving target, namely, a larger spacing results in greater errors, especially in low SNR conditions. When SNR $\geqslant 0$ dB, the absolute angular errors were less than $1°$, and as the SNR increased, the absolute errors tended to stabilize. Therefore, noise is a source of uncertainty, and in order to ensure the effective application of our method, SNR $\geqslant 0$ dB is required in this scenario. The absolute target angular errors, obtained through the sequential measurement procedure and divided by the fixed angle and time, are shown in Fig. 12.

In real-world scenarios, sequential measurements are based on fixed time intervals, rather than on fixed angles. The angle-estimation results, as shown in Fig. 12, obtained using the two partitioning methods confirm that the angle results obtained at fixed time and fixed angle have the same accuracy and validate the consistency between the theory and actual measurements. This provides a basis for selecting a fixed time in Algorithm 1 and establishes a solid foundation for the practical application.

However, the angle unit may deviate due to calculation errors when estimating the target angle. Suppose that the FFT number is 512 when $d = \lambda$, and the absolute angular error owing to the deviation is $0.11°$, the estimated target angle was either $0°$ or $\pm 0.11°$. Nevertheless, we conclude that the absolute angular errors of a target at any position and moving in any direction using a fixed angle division are equivalent to those obtained using a fixed time division. Therefore, the nonuniqueness in angle shifts encountered when handling angle ambiguity can be effectively addressed through the sequential measurement procedure and voting mechanism, thereby ensuring more stable and reliable angle estimates.

B. Experimental Setup and Results

In the experiments, we used an MMW radar sensor (AWR2243BOOST) manufactured by Texas Instruments (Dal-

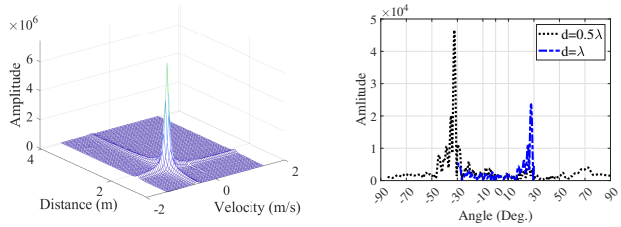
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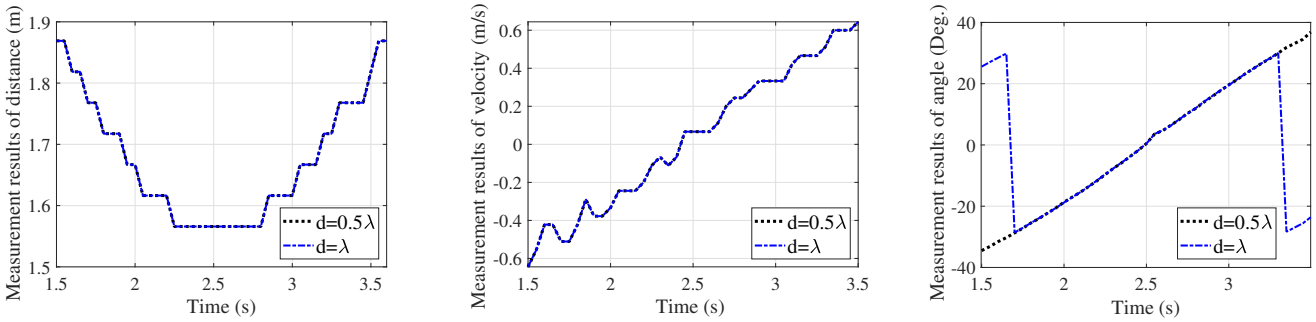
(a) (b)

Fig. 13. Slideway setup employed in the experiment. (a) Photo. (b) Illustration.



(a) (b)

Fig. 14. Measurement results for the 30th frame (moving corner reflector along a slideway). (a) Range-Doppler (RD) map. (b) Angular spectra.



(a) (b) (c)

Fig. 15. Measurement results of multiple frames (moving corner reflector along a slideway). (a) Distance. (b) Radial velocity. (c) Angle of the target.

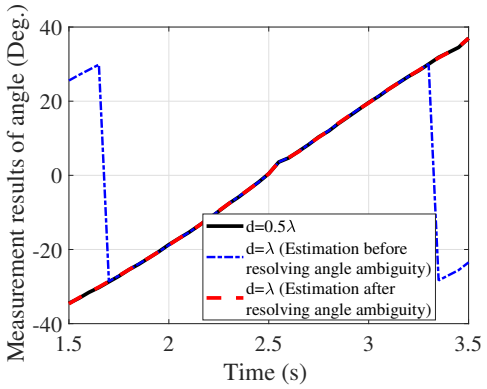


Fig. 16. Measurement results of angle versus time (moving corner reflector along a slideway).

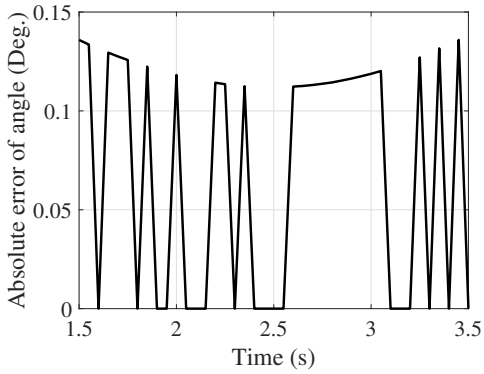


Fig. 17. Absolute angular errors between the full ULA and SULA results for solving angle ambiguity versus time (moving corner reflector along a slideway).

las, TX, USA) [33] with the parameter settings in Table I. It comprises MMWCAS-DPS and MMWCAS-RF evaluation modules, four AWRx chips, and an embedded antenna array, comprising 86 antenna elements with half-wavelength spacing. In our experiments, the first 80 elements were used for validation. For $d = \lambda$ configuration, odd-indexed elements within the first 80 channels (i.e., indices $1, 3, 5, ..., 79$) were programmatically selected through sparse sampling. The number of frames transmitted by the radar was 100. We considered a moving corner reflector along a slideway and a moving pedestrian as the targets, and verified that the proposed method can effectively address angle-ambiguity.

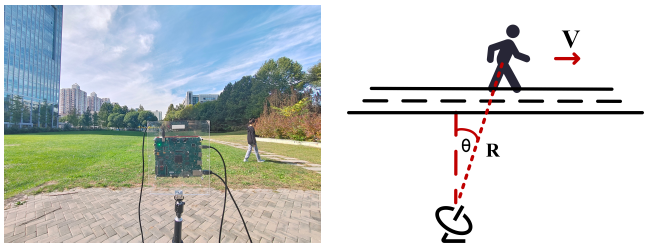
1) Case 1: Moving Corner Reflector Along a Slideway: The image and illustrations of a moving corner reflector along a slideway is shown in Fig. 13.

The slideway was placed perpendicular to the radar normal. The length of the slideway was $4\,m$ and the target velocity was $1\,m/s$. We selected 30-70th frames for the analysis, which represented a stable stage of the target motion. The measurement results for one frame are shown in Fig. 14, and the results of multiple frames are shown in Fig. 15.

The RD map of the $30\mathrm{th}$ frame shown in Fig. 14(a) indicates the estimated distance and radial velocity, which are consistent with the results of our previous analysis, indicating that the grating lobes exclusively impact the target angle. Fig. 14(b) shows the angular spectra of the $30\mathrm{th}$ frame for $d = 0.5\lambda$ and $d = \lambda$, corresponding to the cases of angle unambiguity and ambiguity, respectively.

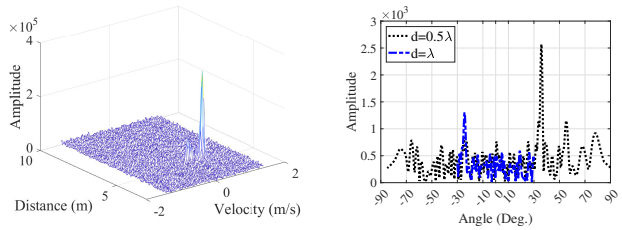
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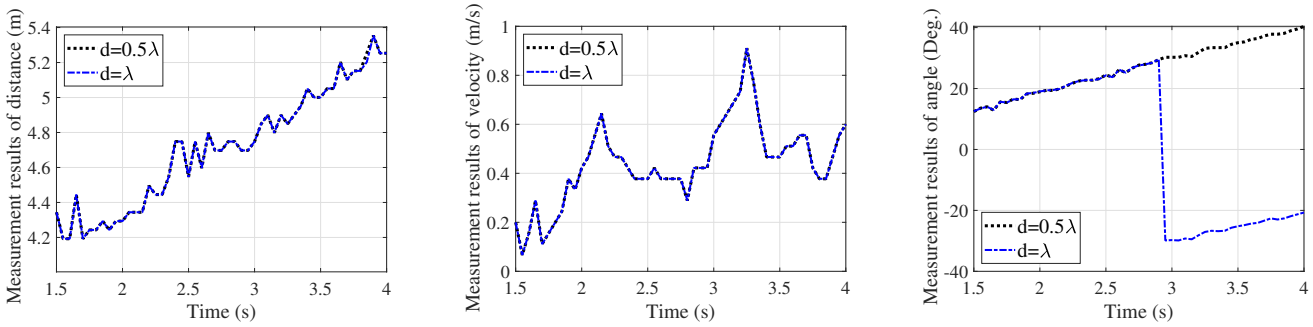
(a) (b)

Fig. 18. Pedestrian scene employed in the experiment. (a) Photo. (b) Illustration.



(a) (b)

Fig. 19. Measurement results for the 70th frame (pedestrian). (a) RD map. (b) Angular spectra.



(a) (b) (c)

Fig. 20. Measurement results for multiple frames (pedestrian). (a) Distance. (b) Radial velocity. (c) Angle of the target.

In Fig. 15, the horizontal axes represent the time from the 30th to the 70th frame, and the vertical axes represent the measurement results. The estimated target distances and radial velocities for $d = 0.5\lambda$ and $d = \lambda$ were identical, indicating they remained unaffected by the grating lobes, which is consistent with the analysis presented in Section III. Additionally, the signals from the 30th to 70th frames correspond to the target motion time from 1.5-3.5 s. Furthermore, the results are symmetrical around the 50th frame, i.e., 2.5 s.

The measurement results of target angle versus time for different cases are shown in Fig. 16. Compared to Fig. 15(c), Fig. 16 includes the angle estimations obtained after solving the angle ambiguity. The results are similar to those of the full ULA and correspond to Fig. 2. The absolute errors between the full ULA results and those obtained by SULA after solving the angle ambiguity are shown in Fig. 17, all of which are less than 0.15°. The absolute angular error increased as the target moved farther from the radar normal and vice versa. However, the overall trend is symmetrical, corresponding to the results shown in Fig. 9. Additionally, deviations in the angle units during processing may cause the absolute error to fluctuate; however, overall, a symmetrical trend can be clearly observed.

2) Case 2: Pedestrian Target (Angle Between Movement and Radar Normal is 90°): The pedestrian scene employed for the experiment is shown in Fig. 18, wherein the pedestrian is moving in the positive direction. According to a survey, the average walking velocity of pedestrians is approximately 1.0-1.1 m/s [34]; therefore, we selected frames 30-80th frames for the analysis.

The measurements of one frame are shown in Fig. 19, and the results of multiple frames are shown in Fig. 20. Fig. 19(a) depicts the RD map of the 70th frame, whereas Fig. 19(b) illustrates the angle estimates of the 70th frame for $d = 0.5\lambda$ and $d = \lambda$, corresponding to the cases of angle unambiguity and ambiguity, respectively.

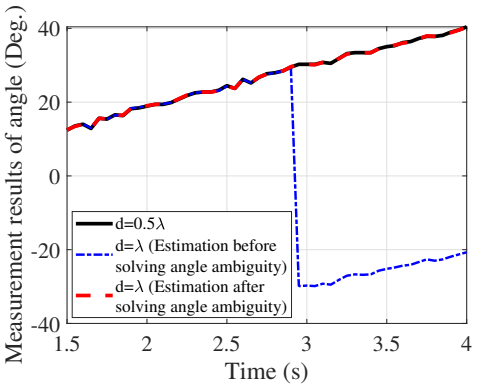


Fig. 21. Measurement results of angle versus time (pedestrian).

In Fig. 20, the horizontal axes represent the time from the 30th to the 80th frames, and the vertical axes represent the measurement results. As the speed of the pedestrian tends to fluctuate over time, fluctuations can be observed in the velocity measurement results, as shown in Fig. 20(b). In Section III, we concluded that the target distance and radial velocity remain unaffected by the grating lobes. The trend observed in Fig. 20(c) is consistent with that shown in Fig. 2. The average walking velocity of pedestrians was calculated to be approximately 1.09 m/s, consistent with that obtained in [34].

The measurement results of the target angle versus time for different cases are shown in Fig. 21. Compared with

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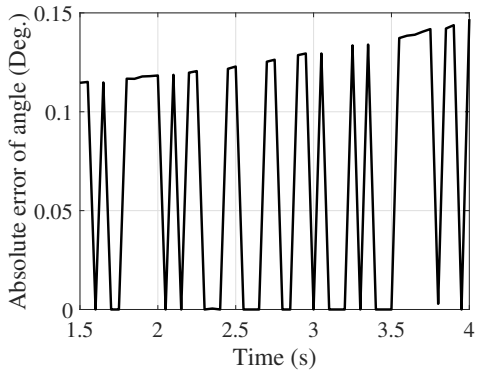


Fig. 22. Absolute angular errors between the full ULA and SULA results for solving angle ambiguity versus time (pedestrian).

Fig. 20(c), Fig. 21 incorporates the results after solving angle ambiguity. Absolute errors of angle between full ULA and SULA results (after addressing angle ambiguity) versus time are shown in Fig. 22. Similarly, we observe that the absolute error of angle estimate is larger when the target is farther from the radar normal and smaller when closer to the normal. Taking the situation with angle ambiguity into consideration, the pedestrian starts with a positive angle. Consequently, the overall error gradually increases, but all errors remain within 0.15°, consistent with the errors observed in Fig. 17.

V. Conclusion

Despite the improvement of angular resolution with a limited number of antenna elements and a simple array structure, SULAs suffer from high grating lobes, which could cause severe angle ambiguity in MMW radar perception. To address the issue, this article proposes a new estimation method of target angles for MMW radars with SULAs. It takes advantage of the restriction among distance, radial velocity and angle measurements for radar targets. In addition, a sequential measurement procedure and voting mechanism is introduced to handle the nonuniqueness in angle shifts in some specific directions of the target. Numerical simulation and experiment results indicate that the proposed method could achieve high-precision angle estimation of radar targets in both ambiguous and unambiguous cases, and adapt to various SULAs with different element spacing. Especially for an SULA with four times the radar wavelength, the estimation error for ambiguous angle measurement is less than 1° when SNR $\geqslant 0$ dB.

Appendix I
Proof of Theorem 1

The proof of Theorem 1 can be divided into the following cases:

(1) Case 1: $\theta_k \in \Omega, \theta_s \notin \Omega$, i.e., $\sin\theta_k \in \left(-\frac{1}{2a}, \frac{1}{2a}\right]$.
Suppose $\theta_s \in \Omega$, the following conditions must then be satisfied:

$$-\frac{1}{2a} < \frac{i}{a} + \sin\theta_k \leq \frac{1}{2a}, i = \pm 1, \pm 2, \ldots \tag{28}$$

Then, (28) can be rewritten as

$$-\frac{1}{2} - a\cdot\sin\theta_k < i \leq \frac{1}{2} - a\cdot\sin\theta_k \tag{29}$$

Since the accepted input range for the arcsine function are real numbers within the range of -1 to 1, we obtain

$$\begin{cases} -\frac{1}{2} - a\cdot\sin\theta_k \in [-1, 0) \\ \frac{1}{2} - a\cdot\sin\theta_k \in [0, 1) \end{cases} \tag{30}$$

Evidently, i has no solution; therefore, the assumption is invalid, i.e., $\theta_k \in \Omega$ and $\theta_s \notin \Omega$ occur simultaneously.

(2) Case 2: $\theta_k \notin \Omega, \theta_s \in \Omega$, i.e., $\sin\theta_s \in \left(-\frac{1}{2a}, \frac{1}{2a}\right]$.
As $\theta_s \in \Omega$, the following conditions must be satisfied:

$$-a\cdot\sin\theta_s \in \left[-\frac{1}{2}, \frac{1}{2}\right) \tag{31}$$

Suppose $A = -\frac{1}{2} - a\cdot\sin\theta_k$ and $B = \frac{1}{2} - a\cdot\sin\theta_k$, which can be regarded as translations by $-a\cdot\sin\theta_s$. Thus, we obtain

$$|B - A| = 1 \tag{32}$$

Evidently, only one solution exists for i; therefore, one main or grating lobe exists within the unambiguous angle range. Thus, the validity of Theorem 1 is verified.

Appendix II
Proof of Theorem 2

Suppose that the true angles of two consecutive measurements are θ_1 and θ_2, and their estimates are

$$\hat{\theta}_1 = \sin^{-1}\left(\frac{i}{a} + \sin\theta_1\right), i \in \left(-2a, 2a\right) \tag{33}$$

$$\hat{\theta}_2 = \sin^{-1}\left(\frac{j}{a} + \sin\theta_2\right), j \in \left(-2a, 2a\right) \tag{34}$$

where i and j are integers. Additionally, the angle shifts corresponding to all possible paths can be expressed as

$$\Delta\theta_x = \sin^{-1}\left(\frac{j}{a} + \sin\theta_2\right) - \sin^{-1}\left(\frac{i}{a} + \sin\theta_1\right), x \in \left[1, 4a^2\right] \tag{35}$$

In this case, the correct angle shift is expressed as

$$\Delta\theta = \theta_2 - \theta_1 \tag{36}$$

Our objective is to establish the number of solutions for $\Delta\theta_x = \Delta\theta$, i.e., $\Delta\theta_x - \Delta\theta = 0$, which can be expressed as

$$\sin^{-1}\left(\frac{j}{a} + \sin\theta_2\right) - \sin^{-1}\left(\frac{i}{a} + \sin\theta_1\right) - (\theta_2 - \theta_1) = 0 \tag{37}$$

Eq. (37) can be rewritten as

$$\sin^{-1}\left(\frac{j}{a} + \sin\theta_2\right) - \theta_2 - \left[\sin^{-1}\left(\frac{i}{a} + \sin\theta_1\right) - \theta_1\right] = 0 \tag{38}$$

Subsequently, we construct the following function:

$$f\left(x\right) = \sin^{-1}\left(c + \sin x\right) - x \tag{39}$$

The derivative of (39) can be expressed as

$$f'\left(x\right) = \frac{\cos x}{\sqrt{1 - \left(c + \sin x\right)^2}} - 1 = \frac{\cos x - \sqrt{1 - \left(c + \sin x\right)^2}}{\sqrt{1 - \left(c + \sin x\right)^2}}, \tag{40}$$

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Since $\sqrt{1-(c+\sin x)^2} > 0$, the comparison between $f'(x)$ and 0 can be changed to a comparison between $\cos x - \sqrt{1-(c+\sin x)^2}$ and 0. Additionally, because $\cos x > 0$ and $\sqrt{1-(c+\sin x)^2} > 0$, the comparison between $\cos x - \sqrt{1-(c+\sin x)^2}$ and 0 can be changed to a comparison between $(\cos x)^2 - \left(\sqrt{1-(c+\sin x)^2}\right)^2$ and 0.

Thus, the monotonicity of function $f(x)$ can be characterized as follows:

$$\begin{aligned}
&(\cos x)^2 - \left(\sqrt{1-(c+\sin x)^2}\right)^2 \\
&= \cos^2 x - \left(1-(c+\sin x)^2\right) \\
&= \cos^2 x - \left(1-c^2-2c\sin x-\sin^2 x\right) \\
&= c\left(c+2\sin x\right)
\end{aligned} \tag{41}$$

(1) Case 1: $c > 0$.
When $\sin x < -\frac{c}{2}$, $c+2\sin x < 0, f'(x) < 0$. When $\sin x > -\frac{c}{2}$, $c+2\sin x > 0, f'(x) > 0$.
Therefore, $f(x)$ initially decreases and then increases.

(2) Case 2: $c < 0$.
When $\sin x < -\frac{c}{2}$, $c+2\sin x < 0, f'(x) > 0$. When $\sin x > -\frac{c}{2}$, $c+2\sin x > 0, f'(x) < 0$.
Therefore, $f(x)$ initially increases and then decreases.

(3) Case 3: $c = 0$.
$f'(x) = 0$. Consequently, $f(x)$ remains constant.

Based on the above analyses, we considered the following functions:

$$f_1(\theta_1) = \sin^{-1}\left(\frac{i}{a}+\sin\theta_1\right) - \theta_1 \tag{42}$$

$$f_2(\theta_2) = \sin^{-1}\left(\frac{j}{a}+\sin\theta_2\right) - \theta_2 \tag{43}$$

As they share the same monotonicity, we obtain

$$f_2(\theta_2) = f_1(\theta_1) \tag{44}$$

In summary, the intersection points of the two functions are solutions to the nonunique problem of target-angle shifts. Consequently, Theorem 2 was validated.

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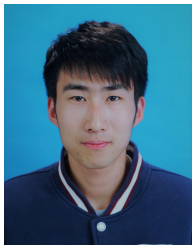
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